

Collapse-resistant fully neural MS2LDA on MSnLib: a reproducible applied-metabolomics checkpoint

MS2LDA research checkpoint

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Abstract

Classical mass-spectrometry latent Dirichlet allocation (MS2LDA) discovers reusable Mass2Motifs from co-occurring fragments and neutral losses, but its iterative inference limits amortized deployment. We evaluated whether discovery itself can be made fully neural without a Tomotopy teacher, DreaMS embeddings, variational Bayes (VB), or chemistry labels and test information in the frozen run’s parameter training. A direct spectrum-level neural topic model collapsed; sparse token routing with balanced Sinkhorn assignments and deterministic topic recycling was associated with recovery from gross usage collapse, while an embedding-regularized neural topic model (ERNTM) constraint reduced semantic duplication. The historical epoch-36 checkpoint replayed exactly. A fresh fixed-protocol run, performed without retuning, selected epoch 38 and obtained held-out negative log-likelihood (NLL) 8.3563, top-word diversity 0.7354, mean normalized pointwise mutual information (NPMI) -0.5699 , 132 corpus-active topics, and 231.5 mass-99 distinct-topic equivalents. Its cached one-pass inference reached 5,529.0 spectra/s. Tomotopy $K=1000$ remained stronger on NPMI (-0.2996), diversity (0.8145), active topics (363), and chemical coverage. The result is therefore a working, fast, collapse-resistant fully neural research model, although it does not yet achieve Tomotopy parity.

1 Introduction

MS2LDA transfers the document–topic–word abstraction to tandem mass spectrometry (MS/MS): spectra are documents, fragments and neutral losses are words, and recurring co-occurrence patterns are Mass2Motifs. These motifs can reflect shared substructures and support unsupervised exploration beyond direct library matching [1, 2]. The classical approach is well established, but both training and held-out inference are iterative.

Neural topic models promise fast amortized inference, flexible conditioning, and differentiable integration with downstream models. They also have a known failure mode: several nominal topics can converge to repetitive semantic regions or receive negligible corpus mass [3]. Broader evaluations have found that neural topic models can be less stable and less aligned with applied categories than classical LDA [5]. This is especially relevant for MS/MS, where documents are short and sparse, frequent peaks dominate reconstruction gradients, rare motifs can still be chemically valuable, and a large requested topic count K creates many weakly supervised components.

This work asks a bounded applied question: can published neural topic-model regularization, combined with MS/MS-specific sparse routing, yield a useful fully neural discovery model? It does not seek a new general machine-learning algorithm. The contribution is a self-contained implementation, a leakage-controlled benchmark, an exact replayable model bundle, and a candid comparison with Tomotopy.

2 Methods

2.1 Data source and spectrum representation

We used the public positive-ion-mode MSnLib assets in Zenodo record 20179680 (<https://zenodo.org/records/20179680>). The acquisition script downloads only the two required archives, resumes interrupted transfers, rejects unsafe archive entries, and verifies archive size, MD5 and SHA-256 digests, ZIP integrity, and the hashes of extracted inputs. The primary input is `Corinna_Library_filtered_positive.mgf` (SHA-256 `c1a2389...714682f`), where MGF means Mascot Generic Format and each block represents one tandem mass spectrum. A valid block requires a Universal Spectrum Identifier (USI), feature identifier, Simplified Molecular Input Line Entry System (SMILES) string, International Chemical Identifier key (InChIKey), and precursor mass-to-charge ratio (m/z).

Preprocessing validates finite positive peak m/z and non-negative intensity, normalizes each original spectrum to maximum intensity one, retains peaks with $0 < m/z \leq 2000$ and relative intensity 0.01–1.0, keeps at most the 1,000 strongest peaks, and rejects spectra with fewer than five retained peaks. This retained 38,888 of 41,568 input blocks; 2,680 failed the declared parsing, metadata, structure, or peak filters. A retained peak at $m/z = 123.45$ becomes the word `frag@123.45`. When precursor minus fragment mass exceeds 0.01, the same peak also yields a neutral-loss word such as `loss@18.01`. Masses are rounded to two decimal places. Relative intensity is represented as an integer count: both words are repeated $\text{round}(100 \times \text{relative intensity})$ times. In the sparse matrix this is stored as one nonzero vocabulary entry plus its count, not as materialized duplicate tokens. A fragment and its derived loss form one *physical peak group*, which is always masked atomically to avoid leaking the same measured peak across input and prediction target.

2.2 Structural split, vocabulary, and held-out completion

The RDKit cheminformatics toolkit parses each SMILES string and checks that its derived connectivity block agrees with the supplied InChIKey. Cyclic molecules are grouped by chirality-free Bemis–Murcko scaffold (their ring-and-linker core); acyclic molecules are grouped by connectivity identity. Groups are ordered by decreasing size, with a seed-42 hash tie-break, and greedily placed to minimize normalized overflow relative to 70/10/20 percent targets. All spectra from a group remain together. The resulting split contains 27,222 training, 3,889 validation, and 7,777 test spectra. An audit found zero overlap of both compound connectivity keys and structural split groups.

Structural metadata is therefore used to create and audit the split, but no SMILES, InChIKey, scaffold, chemical label, or library annotation is supplied to the neural model or its objective. The 21,233-word vocabulary is then built only from training spectra: words must occur in at least three training documents, no corpus-frequency or top-frequency filter is applied, and column order follows first appearance in the raw training spectra. A clean rebuild reproduced this order exactly.

Validation and test use deterministic document completion. A seed-42 hash of the spectrum identity and original peak index divides whole peak groups into 50 percent observed and 50 percent completion partitions, retaining at least one group on each side. The observed half is renormalized using only its own maximum, so even the held-out maximum cannot leak into inference. The fresh completion target retains its full-spectrum count weights. The fresh test NLL denominator contains 2,534,703 in-vocabulary completion-token copies from 7,776 of 7,777 spectra. A further 70,016 copies (2.688 percent) are out of vocabulary (OOV); they are reported but excluded from NLL.

2.3 Neural architecture

The model is a neural token-assignment topic model, rather than a conventional variational autoencoder (VAE). It retains the MS2LDA admixture interpretation—each spectrum is a mixture of topics and each topic is a distribution over fragment and neutral-loss words—but replaces per-spectrum iterative inference with one learned routing pass. There is no Gaussian latent variable, Kullback–Leibler (KL) divergence-to-prior term, Tomotopy teacher, or local VB refinement.

For vocabulary word w , the fixed input feature $f_w \in \mathbb{R}^{64}$ concatenates a 48-dimensional skip-gram-with-negative-sampling (SGNS) embedding learned only from training co-occurrences, 14 sine/cosine Fourier mass features at seven frequencies, and a two-dimensional fragment/loss indicator. A learned bias-free linear projection maps this vector to a normalized 128-dimensional token embedding e_w . Prototype initialization is also data-only: weighted k-means++ chooses 500 projected training-vocabulary tokens, using square-root corpus frequency times squared inverse document frequency as the sampling weight, but performs no Tomotopy initialization or Lloyd updates.

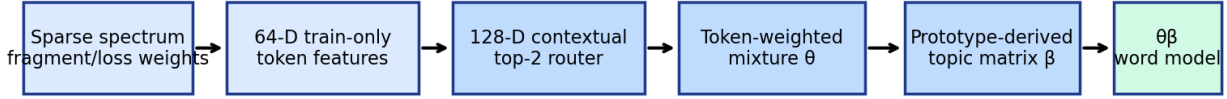
Routing is contextual at the level of an individual observed token. For token i in spectrum d , the model first forms the weighted mean c_{di} of all other token embeddings in that spectrum. A two-layer 256-unit multilayer perceptron (MLP) predicts a correction to the token embedding,

$$r_{di} = \text{norm}[e_{w_i} + g(e_{w_i}, c_{di})], \quad \ell_{dik} = r_{di}^T \bar{t}_k,$$

where t_k is a learned 128-dimensional topic prototype. Softmax logits use a temperature annealed from 0.5 to 0.1. Only the two largest probabilities are retained and renormalized; a straight-through estimator supplies gradients through this discrete top-2 choice during training. The spectrum mixture is then the normalized, token-weighted sum of those sparse assignments,

$$\theta_{dk} = \frac{\sum_{i \in d} x_{di} q_{dik}}{\sum_{i \in d} x_{di}},$$

One-pass inference and shared topic geometry



No local optimization: θ is produced directly; the same prototypes route tokens and define β

Training-only collapse controls with distinct roles

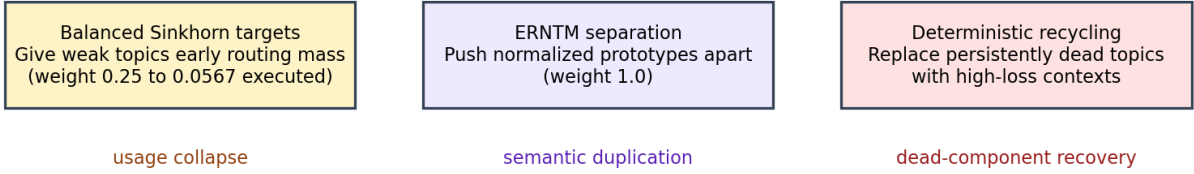


Figure 1: Supported fully neural discovery architecture and the three training-only anti-collapse controls. Tomotopy and chemistry enter only after neural training has finished.

so the spectrum–topic mixture θ for a complete spectrum requires one router evaluation rather than an optimization loop.

The decoder shares the same geometry instead of learning an unrelated dense matrix. The topic–word matrix β and the MS2LDA admixture likelihood are

$$\beta_{kw} = \text{softmax}_w \left(2 \bar{t}_k^T \bar{e}_w / 0.2 \right), \quad p(w | d) = \sum_k \theta_{dk} \beta_{kw}.$$

Consequently each learned prototype has two linked roles: it attracts routed tokens to produce θ , and its similarity to every vocabulary embedding defines the corresponding Mass2Motif row of β . Figure 1 separates this one-pass inference path from the training-only controls.

2.4 Training-only feature learning and initialization

SGNS is trained before the topic model, using only training documents. It runs for five epochs, draws eight positive co-occurrence pairs per document per epoch in proportion to the intensity-derived counts, and contrasts each with five negative words sampled according to corpus frequency raised to 0.75. SparseAdam uses batch size 8,192 and learning rate 0.01. Averaged, normalized source/context vectors form the 48 SGNS coordinates. These are concatenated with 14 Fourier mass coordinates (sine and cosine at frequencies 1, 2, 4, 8, 16, 32, and 64 after scaling mass by 1,000) and a two-value fragment/loss indicator, then normalized to yield the fixed 64-dimensional feature.

The learned $64 \rightarrow 128$ projection is initialized orthogonally. Topic prototypes are seeded by weighted k-means++ sampling without subsequent Lloyd refinement; the weight is square-root corpus frequency times squared inverse document frequency (IDF). Thus initialization uses spectral co-occurrence, mass, type, and training frequency only. It does not copy topics from Tomotopy or the published MSnLib motif set.

2.5 Anti-collapse mechanism and optimization

2.5.1 Two failure modes

Collapse has two distinct forms here. In *usage collapse*, most observed mass is routed through a small subset of prototypes and the others receive almost no learning signal. In *semantic collapse*, several active prototypes occupy the same region and yield near-duplicate top words. The training objective addresses these failure modes separately rather than treating reconstruction loss as sufficient.

2.5.2 Corrupted views and router objective

Each epoch uses one of four deterministic pairs of corrupted training views; each side independently retains 80% of physical peak groups, with a fragment and its derived loss kept together, and the pair rotates by epoch. Because the views can overlap, this is not strict complementary masking; it reduces identity shortcuts and trains robustness to missing peaks. Each retained view is renormalized to its own maximum. In the router block, the decoder probabilities β are cached and detached once per epoch and the mixture from either view must predict the tokens in the other:

$$\mathcal{L}_{\text{router}} = \frac{1}{2} [\text{NLL}(B | \theta^A, \beta) + \text{NLL}(A | \theta^B, \beta)] + \lambda_{\text{bal}} \mathcal{L}_{\text{Sinkhorn}} + 0.1 \text{JS}(\theta^A, \theta^B),$$

where JS is Jensen–Shannon divergence. Shared token projections and prototypes can still change because they define routing, so “cached decoder” does not mean that all decoder geometry is immutable for the epoch. Cross-view completion discourages reliance only on a token’s own identity, while JS consistency asks the two partial views to describe the same spectrum similarly. The routed token itself is excluded when its contextual mean is formed.

2.5.3 Balanced Sinkhorn usage target

The first anti-collapse control is a stop-gradient balanced Sinkhorn target. Five log-space normalization iterations convert the token–prototype logits in each mini-batch into a transport plan whose rows sum to one and whose aggregate columns are equal. Cross-entropy trains the ordinary router probabilities toward this plan. This is an early training scaffold, not a claim that chemical motifs are uniformly prevalent: its weight is held at 0.25 for the first ten epochs and then reduced linearly toward 0.05 at epoch index 40. The executed run used zero-based indices 0–39, so its 40th and final epoch used weight 0.0567. This documents the frozen execution rather than silently changing it. The schedule gives initially weak prototypes enough routed examples to start learning while leaving reconstruction increasingly responsible for final prevalence.

2.5.4 ERNTM prototype separation

The second control is the ERNTM prototype-separation constraint [4]. For normalized prototypes, $S_{kj} = \bar{t}_k^T \bar{t}_j$, and the model minimizes row-wise cross-entropy with the identity matrix as the target. Each prototype must therefore be more similar to itself than to any competing prototype. This directly penalizes semantic duplication but does not by itself guarantee that a topic is used. It is applied with weight 1.0 during four topic/decoder updates per epoch. Those updates hold the routed assignments fixed and optimize cross-view completion plus a weight-0.25 local decoder term that asks the selected topic rows of β to emit their assigned tokens.

2.5.5 Deterministic topic recycling

The third control is deterministic topic recycling. At validation every two epochs, a topic is marked underused when its corpus assignment mass is less than $0.1/K$. If this persists for three consecutive validations, its prototype is replaced by a normalized routing context drawn from a bounded queue of the highest reconstruction-loss tokens. The corresponding AdamW moment rows are reset so stale optimizer state cannot pull the replacement back. A topic can be recycled at most twice. Recycling is designed to restore learning opportunity for a dead component; Sinkhorn supplies early usage pressure; and ERNTM encourages surviving components to remain apart. Their combination coincided with avoidance of gross collapse, but the frozen evidence does not isolate each mechanism in a factorial ablation. None is treated as evidence of chemical validity, which is measured only after training.

2.5.6 Alternating optimization and checkpoint selection

Optimization alternates one complete shuffled pass through all training spectra for the router (batch size 128) with only four topic/decoder mini-batch updates (batch size 512), not four further full passes. Router updates use cached β ; topic updates detach assignments and update the shared topic/decoder geometry. AdamW, an adaptive optimizer with decoupled weight decay, uses learning rate 0.001, weight decay 10^{-4} , and gradient-norm clipping at 5.0. Four central processing unit (CPU) threads are pinned. Training ends at 40 epochs and evaluates validation every two epochs. Validation has two declared roles: it selects the checkpoint and supplies corpus-usage estimates for recycling. A validation checkpoint is saved before any recycling decision made at that validation. Selection maximizes validation mass-99 distinct-topic equivalents, breaking ties with lower completion NLL and then the earlier epoch. Test spectra and all chemistry are opened only after the within-run checkpoint choice is frozen.

2.6 Comparator and evaluation

Tomotopy is the optimized software implementation of classical LDA used by MS2LDA. The comparator uses $K = 1000$, $\alpha = 0.6$, $\eta = 0.1$, seed 42, and the same prepared training documents and 21,233-word vocabulary. It trains in blocks of 50 iterations with six workers (parallel setting 3), stopping at 5,000 iterations or after

Table 1: Definitions for non-chemical evaluation. All word co-occurrence statistics are computed from the 27,222 training documents.

Metric	Definition and denominator	Interpretation
Completion NLL	Mean negative natural log probability per in-vocabulary completion-token copy, with θ inferred from the observed half. OOV copies are excluded.	Lower is better predictive fit.
Top-10 word diversity	Unique vocabulary words across all $K \times 10$ top-word slots, divided by $K \times 10$.	Higher means less repetition, not necessarily chemical coherence.
NPMI	Normalized pointwise mutual information for all 45 pairs among each topic’s top ten words, using binary training-document presence; unseen pairs score -1 .	Range $[-1, 1]$; higher is more co-occurrence.
Corpus-active topics	Topics whose mean mixture weight over observed test halves is at least $1/K$ (0.002 neural; 0.001 Tomotopy).	Diagnostic only; higher is not automatically better.
Effective topic count	$\exp[-\sum_k \theta_k \log \theta_k]$ for one spectrum.	Mixture breadth; one for a pure spectrum.
Mass-99 distinct-topic equivalents	Take the smallest usage-ranked set carrying at least 99 percent of corpus mass; count unique words in its top-ten lists and divide by ten.	Redundancy-adjusted inventory diagnostic, not a count of chemically validated motifs.
Cached throughput	Median spectra/s over five warmed repetitions on 128 prepared spectra; file input and preprocessing are excluded.	Higher is faster for this declared setup.

the relative perplexity change remains below 0.005 for ten checks. Held-out inference uses 100 iterations and one worker. Tomotopy is never an initializer or teacher.

This is an established-setting comparison, not a capacity- or core-matched algorithm comparison: the neural model uses 500 topics and four CPU threads, whereas Tomotopy uses 1,000 topics and one worker for held-out inference. Both see the same spectra, completion masks, vocabulary, and scoring functions, so completion and coherence metrics remain informative. Counts whose threshold depends on K are method-relative.

For context, the neural full-test mixtures have median effective topic count 14.75 (mean 17.92; 95th percentile 41.94) and median maximum topic probability 0.203. The neural mass-99 value 231.5 comes from 2,315 unique top-word slots among the 266 topics carrying 99.02 percent of test mass.

2.7 Chemical annotation and evaluation

Chemistry is evaluated only after model and checkpoint selection, using θ inferred from each *full* test spectrum. Automated Mass2Motif Annotation Guidance (MAG) turns each topic’s top 20 β words and weights into a pseudo-spectrum. Spec2Vec, a learned spectral-similarity embedding, embeds that pseudo-spectrum; a FAISS vector-search index returns up to 1,000 candidate neighbors, from which ten unique full-InChIKey molecules are kept. Before index construction, every validation and test connectivity key is removed from the reference library and the retained index is audited for zero leaks. Hits are clustered at cosine similarity 0.9 and passed through motif optimization. A successful annotation is summarized by a MACCS structural-key fingerprint whose bits occur in at least 80 percent of clustered structures.

Annotation coverage is the proportion of all requested topics for which MAG optimization retains at least one fragment or loss. A topic becomes SOS-evaluable only if it also has at least one associated held-out compound with a valid fingerprint. Dominant association assigns each of the 7,777 test spectra to its largest- θ topic. High-confidence association retains each topic-spectrum pair with $\theta_k \geq 0.5$; the reported count is therefore a pair count (an exact 0.5 tie could associate one spectrum twice), not a guaranteed count of unique spectra.

The structure overlap score (SOS) is asymmetric annotation containment,

$$\text{SOS}(a, m) = \frac{|\text{bits}(a) \cap \text{bits}(m)|}{|\text{bits}(a)|},$$

where a is the consensus annotation and m a held-out molecule. It ranges from zero to one; higher means that more annotation bits occur in the molecule. Within a topic, spectra are deduplicated by compound connectivity and compounds are averaged equally; the reported global mean then averages eligible topics equally. It is not spectrum-weighted. Coverage and conditional SOS answer different questions and must be reported together.

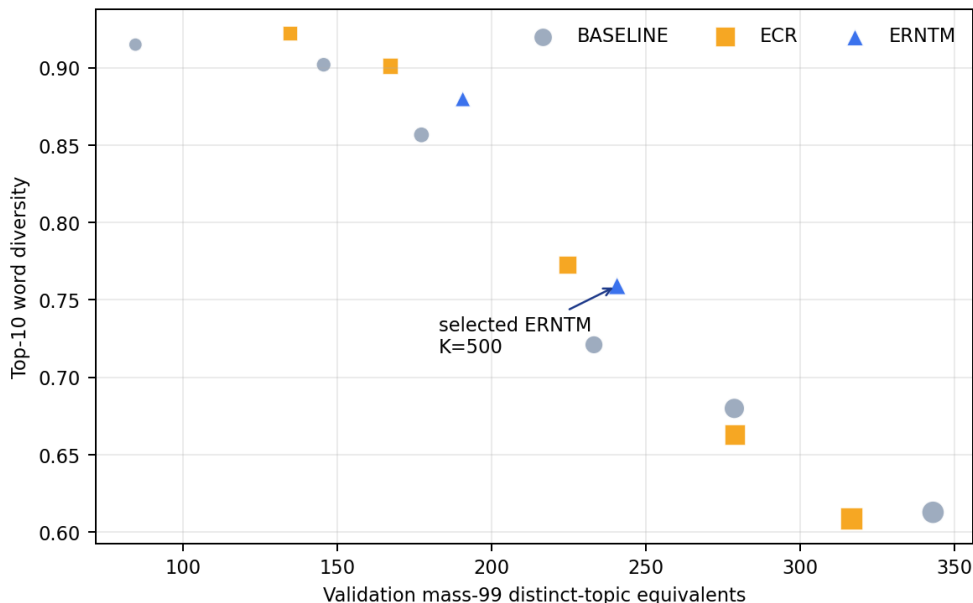


Figure 2: Validation metrics from the exploratory capacity screen. Marker area increases with nominal K ; family selection also considered the exploratory evidence described in the text.

3 Results

3.1 Capacity and model selection

Nominal K was not treated as the number of usable motifs. Across the capacity screen, increasing K expanded the mass-carrying inventory but usually reduced global diversity. Embedding clustering regularization (ECR), adapted from the published ECR topic model (ECRTM) [3], improved selected diversity points but did not dominate the joint likelihood–inventory frontier. The architecture and $K = 500$ choice were exploratory and informed by the earlier study, which included test and chemistry comparisons. Once that family and protocol were frozen, the fresh run selected epoch 38 using validation metrics only; its test results are an end-to-end reproducibility checkpoint, not a new untouched confirmatory study (Figure 2).

3.2 Final non-chemical comparison

The historical epoch-36 checkpoint replayed exactly under the cleaned code: the derived β was bit-for-bit identical, and all headline metrics were reproduced. The fresh unattended 40-epoch run then selected epoch 38 by the frozen validation rule. It reached test NLL 8.3563, diversity 0.7354, NPMI -0.5699 , 132 active topics, and 231.5 mass-99 distinct-topic equivalents. These values are close to the replayed checkpoint (8.3694, 0.7592, -0.5359 , 140, and 238.7 respectively) and preserve every central conclusion without retuning. The neural model improved completion NLL relative to Tomotopy and was 387-fold faster in the declared cached benchmark. This is a practical pipeline result, not a same-core algorithmic ratio: neural routing used four CPU threads, while Tomotopy used one worker and 100 inference iterations. Tomotopy retained a larger, more coherent active inventory, although its active threshold is lower because it uses twice as many requested topics (Table 2; Figures 3 and 4).

3.3 Chemical evaluation

MAG produced optimized annotations for 136/500 neural topics (27.2 percent), versus 607/1000 Tomotopy topics (60.7 percent). Under dominant association, 105/500 neural topics (21.0 percent) were SOS-evaluable with conditional mean SOS 0.6129, compared with 598/1000 (59.8 percent) at 0.6238. At probability at least 0.5, 26/500 neural topics (5.2 percent) and 106 topic–spectrum associations were evaluable, compared with 333/1000 (33.3 percent) and 961 associations for Tomotopy. The similar dominant conditional means show that evaluable neural annotations have comparable annotation containment under this procedure; without a shuffled or null baseline they do not establish absolute chemical validity. Coverage remains the central weakness (Figure 5).

Table 2: Fresh fixed-protocol test and chemistry comparison, generated from committed JSON evidence. Requested capacity differs, so coverage proportions and method-relative active-topic counts accompany raw counts.

Metric	Neural ERNTM	Tomotopy
Topics requested	500	1000
Held-out NLL (lower is better)	8.3563	9.7569
Top-word diversity (higher is less repetition)	0.7354	0.8145
Mean NPMI (higher is better)	-0.5699	-0.2996
Corpus-active topics (method-relative)	132	363
Annotation coverage	27.2% (136/500)	60.7% (607/1000)
Dominant SOS-evaluable coverage	21.0% (105/500)	59.8% (598/1000)
Dominant mean SOS	0.6129	0.6238
High-confidence topic-spectrum associations	106	961

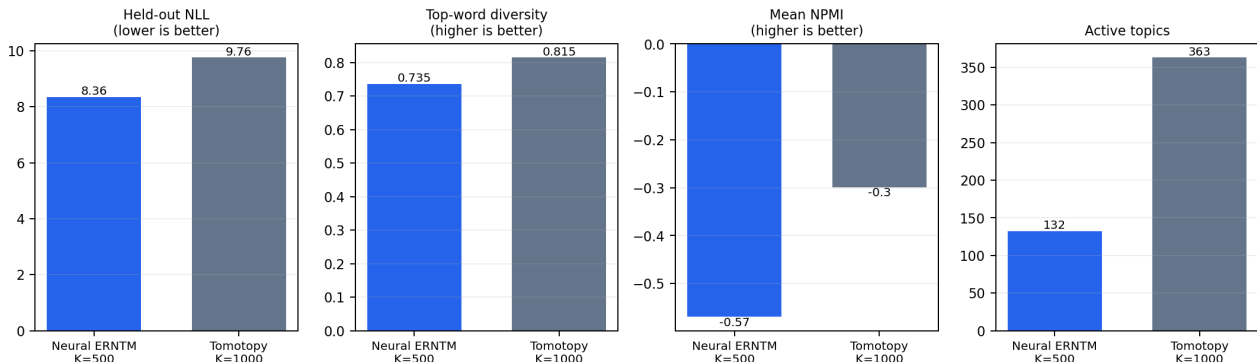


Figure 3: Final held-out comparison. Better NLL does not imply better topic coherence or a larger usable motif inventory.

4 Discussion

4.1 Why fully neural MS2LDA is difficult

The probabilistic and neural models share the same admixture likelihood; the difference is how learning credit is assigned. Classical LDA repeatedly forms token-to-topic expected counts, so a rare but consistent pattern can continue to accumulate evidence. A gradient model can reduce reconstruction loss early with a few broad components. Those components then receive more gradient, while unused prototypes receive little signal and either die or become duplicates. Avoiding a constant encoder (posterior collapse) is therefore not enough: the global topic inventory can still undergo component collapse.

MS/MS intensifies this feedback because spectra are short relative to a 21,233-token vocabulary, common fragments and losses dominate token mass, chemically valuable motifs can be rare, and useful per-spectrum mixtures must be sparse even though the corpus-level inventory should be broad. Uniform use is not chemically justified, yet unconstrained use collapses. The present combination of balanced routing, recycling, and ERNTM separation targets different parts of this tension. The combined model avoided gross collapse and reduced duplication relative to earlier direct neural attempts, but the evidence does not assign causal credit to each mechanism separately, and the model does not reproduce the coherent long tail obtained by the mature classical sampler.

4.2 What ERNTM solved and what remains

The model moved from the earlier K=1000 neural-assignment result—257 active topics but diversity 0.3646—to fresh K=500 results of diversity 0.7354 and 231.5 mass-99 distinct-topic equivalents (0.7592 and 238.7 in the exact replay). This is substantive structural progress, not just a relaxed threshold. Nevertheless, NPMI and chemical coverage remain clearly below Tomotopy. The appropriate claim is therefore “working collapse-resistant fully neural discovery,” not “parity.”

DreaMS, a pretrained mass-spectrum representation model, is non-essential to this contribution. The active evidence establishes that the current model reconstructs its vocabulary and all features from raw training

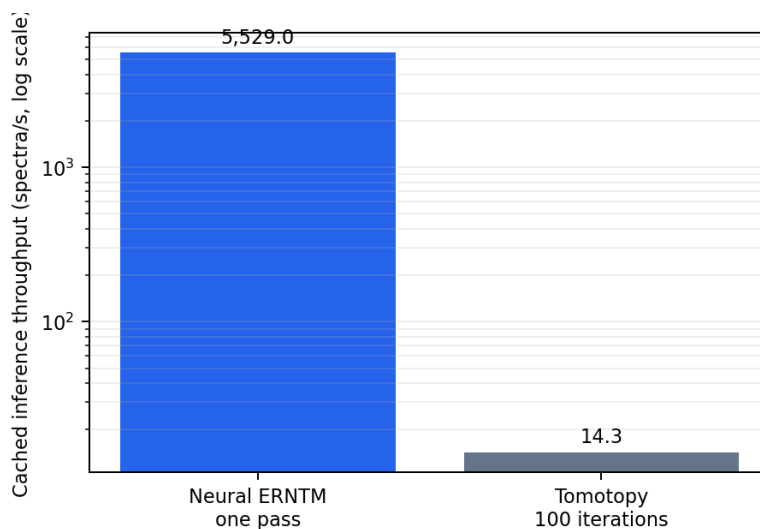


Figure 4: Cached inference throughput on 128 prepared spectra after warm-up, excluding input and preprocessing. Neural uses one routing pass and four CPU threads; Tomotopy uses 100 iterations and one worker.

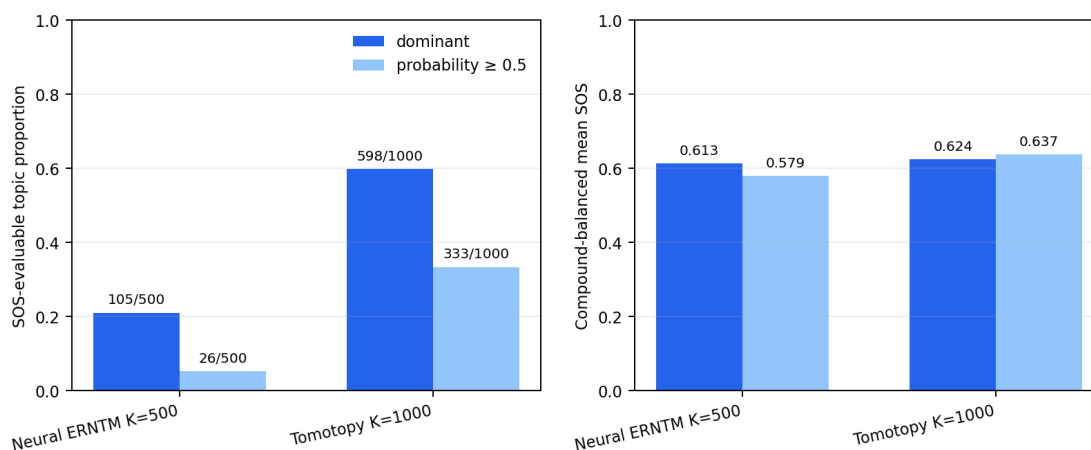


Figure 5: SOS-evaluable proportions (with numerator/denominator labels) and compound-balanced conditional SOS on a zero-to-one axis. Similar conditional means coexist with a large coverage gap.

spectra; it does not attempt to quantify the benefit or cost of DreaMS. Its exclusion reduces dependencies and makes the claim narrower: discovered structure comes from spectral co-occurrence, mass/type features, and the declared neural objectives.

4.3 Applied-metabolomics value

Even without parity, one-pass inference can support high-throughput projection, online screening, differentiable downstream retrieval, and rapid exploratory analysis. The bundled K=500 model is also compact (approximately 6.5 megabytes plus metadata) and directly yields both θ and β . The checkpoint therefore provides a defensible research foundation for improving neural topic quality and for tasks where amortized inference or differentiable integration matters.

5 Limitations

This is a fixed single-seed checkpoint and does not estimate between-seed uncertainty. Capacity and method families were compared on one deterministic split; the test comparisons among validation-selected families are exploratory. NPMI is an imperfect proxy for human or chemical interpretability [5]. MAG coverage depends on the reference library, Spec2Vec, and fingerprint procedure. Throughput is a cached CPU benchmark rather than an end-to-end acquisition latency or a same-core comparison. The committed headline evidence does not preserve neural training wall time, so no training speed claim is made. The machine record identifies the run

as Arm 64-bit (arm64) but does not record a processor model. Finally, the ERNTM constraint is adapted to the MS/MS router rather than claimed as a verbatim reproduction of the text model.

6 Conclusion

A fully neural MS2LDA is possible in the limited but useful sense demonstrated here: it can discover a substantial non-redundant motif inventory without a classical teacher or VB, remain numerically stable, yield evaluable annotations with measurable conditional structure overlap, and infer spectra in one fast pass. It does not yet match Tomotopy’s coherence or chemical coverage. The recommended research direction is to close those gaps while preserving collapse resistance and one-pass inference, using the frozen K=500 ERNTM architecture as the starting point.

A Reproducibility

The public acquisition script verifies Zenodo metadata, archive MD5/SHA-256, ZIP integrity, extraction paths, links, required bytes, and required SHA-256 hashes. The lean neural environment uses Python 3.11, PyTorch 2.6.0, NumPy 1.25.2, SciPy 1.13.1, RDKit 2024.9.6, and Tomotopy 0.13.0. MAG is isolated because its pinned stack differs: Python 3.12.12, Spec2Vec 0.9.1, FAISS-CPU 1.10.0, MatchMS 0.29.0, NumPy 1.26.4, and RDKit 2024.9.6. The acquisition and run commands are deliberately separate; `run` does not download data:

```
conda env create -f environment-neural-ms2lda.yml
conda env create -f environment-msnlib-mag.yml
conda run -n ms2lda-neural python \
  scripts/download_msnlib_validation_assets.py \
  --data-root /path/to/MSnLib-assets
conda run -n ms2lda-neural python -m \
  benchmarks.neural_assignment_ms2lda run \
  --data-root /path/to/MSnLib-assets --run /path/to/run
conda run -n ms2lda-neural python -m \
  benchmarks.neural_assignment_ms2lda status \
  --run /path/to/run
conda run -n ms2lda-neural python -m \
  benchmarks.neural_assignment_ms2lda verify \
  --data-root /path/to/MSnLib-assets --run /path/to/run
```

After initialization, the resumable pipeline executes: prepare and freeze the split and vocabulary; build four paired training views; train SGNS; construct token features and prototypes; train and select the neural checkpoint; evaluate neural test completion; train and evaluate Tomotopy; run neural and Tomotopy chemistry in the isolated MAG environment; generate the report; and verify manifests.

The pipeline records the resolved protocol, code hashes, environment, raw input hashes, split manifest, selected checkpoint hash, and output manifests. It resumes SGNS, neural training, and Tomotopy from verified checkpoints. The committed model bundle contains the fresh validation-selected weights (including token features), vocabulary, protocol, evaluation summaries, chemistry summaries, portable provenance hashes, and a hash manifest. The active weights are the fresh epoch-38 checkpoint. The historical epoch-36 weights are not retained in the active bundle; only their exact replay evidence is committed. The machine-specific run directory and its large θ/β arrays remain external to the repository.

B Unsuccessful approaches

The concise experimental ledger is committed alongside this report. It records random-start amortized collapse, warm-start limitations, hybrid boundaries, the neural K=1000 activity/diversity trade-off, ECR capacity results, and the reason K=500 ERNTM was selected. Implementation code for rejected arms remains in the external all-refs safety bundle and preserved archive branches, not in active main.

C Glossary for newcomers

MS2LDA / LDA Mass-spectrometry latent Dirichlet allocation; LDA is the classical document–topic–word admixture model adapted to spectra.

MS/MS, MSnLib, MGF, and m/z Tandem mass spectrometry; the public multistage mass-spectral library benchmark used here; Mascot Generic Format; and mass-to-charge ratio.

Mass2Motif A reusable distribution over fragment and neutral-loss words, analogous to a topic in text.

K , θ , and β Requested number of topics; a spectrum’s topic-mixture vector; and the topic–word probability matrix.

VB, VAE, and KL Variational Bayes; variational autoencoder; and Kullback–Leibler divergence. None is used for neural spectrum inference here.

SGNS and IDF Skip-gram with negative sampling; inverse document frequency, which upweights words occurring in fewer documents.

MLP, JS, and AdamW Multilayer perceptron; Jensen–Shannon divergence; and the adaptive optimizer with decoupled weight decay used for model training.

Sinkhorn balancing Iterative row/column normalization used to construct a balanced training target across topic prototypes; it is not a claim of uniform chemical prevalence.

ERNTM Embedding-regularized neural topic model. This work adapts its topic-embedding separation constraint, not the entire published text model.

ECR / ECRTM Embedding clustering regularization and its published topic model; an explored alternative, not the selected model.

NLL and NPMI Negative log-likelihood and normalized pointwise mutual information; predictive fit and top-word co-occurrence diagnostics, respectively.

OOV Out of vocabulary: a held-out word absent from the training-only vocabulary.

SMILES and InChIKey Text encodings of molecular structure and a hashed International Chemical Identifier; the connectivity block ignores stereochemical and other suffix information for leakage grouping.

Bemis–Murcko scaffold A molecule’s ring-and-linker core, used to keep structurally related cyclic compounds in one data partition.

MAG, Spec2Vec, FAISS, and MACCS Automated Mass2Motif Annotation Guidance; a learned spectral-similarity embedding; a vector similarity-search library; and a standard structural-key chemical fingerprint family.

SOS Structure overlap score: the fraction of consensus annotation bits contained in a held-out molecule, averaged compound-first and then topic-first.

DreaMS A pretrained mass-spectrum representation model that is not used by this checkpoint.

CPU, CLI, JSON, MD5, and SHA-256 Central processing unit; command-line interface; JavaScript Object Notation; Message Digest 5; and 256-bit Secure Hash Algorithm. The last two identify and verify files rather than model quality.

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